RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

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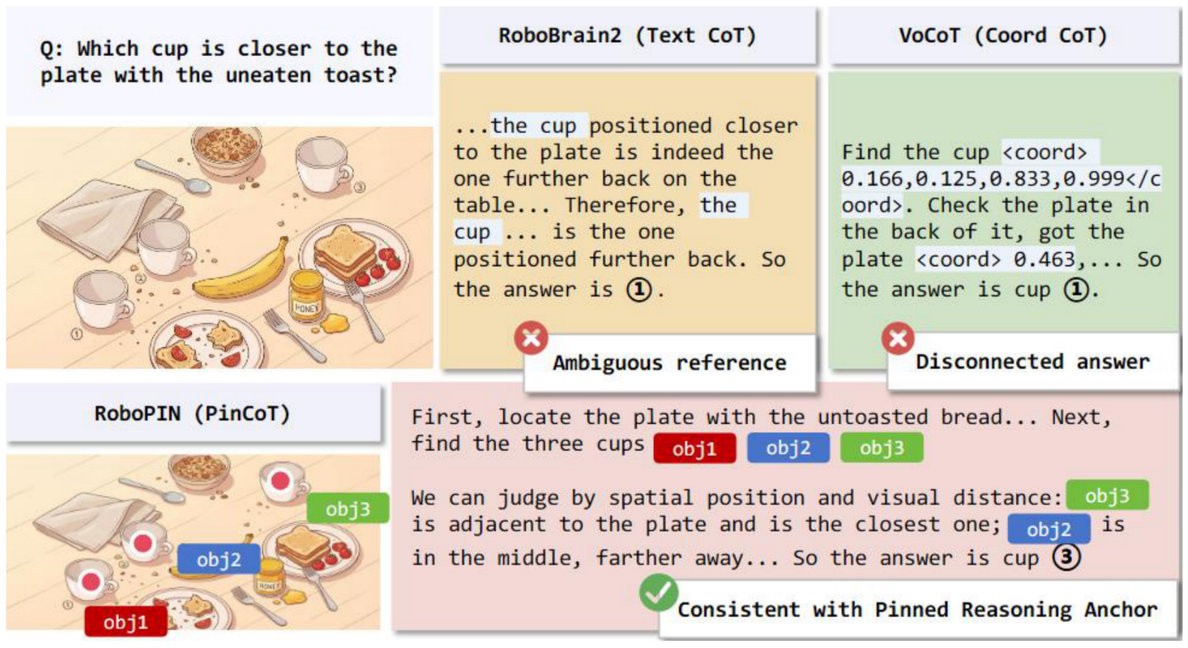
Tianjin University · ACM Multimedia 2026 · Rio de Janeiro, Brazil

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Motivation



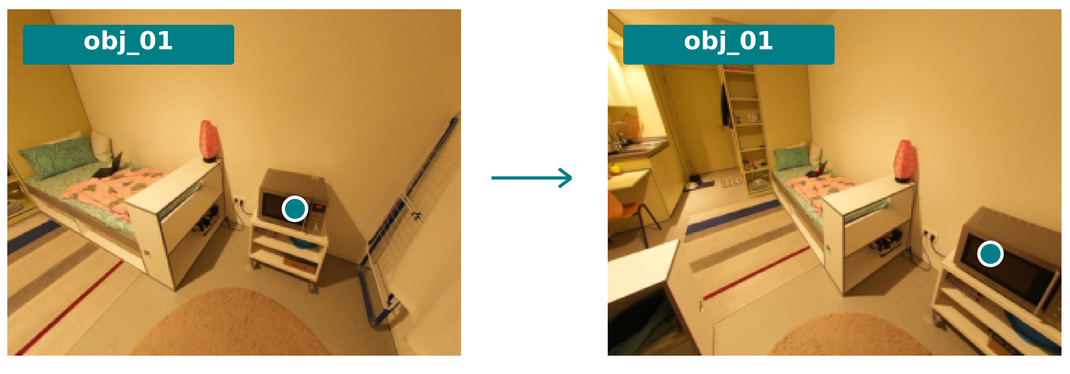
1 Reference drift

Entity references can switch between similar objects across reasoning steps.

2 Reasoning–answer misalignment

The final answer can point to a different entity from the one analyzed.

Key Insight: Persistent Identity



Same entity. Same ID. Different views.

Anchor = name + ID + view + point.

Keep identity stable across steps and viewpoints.

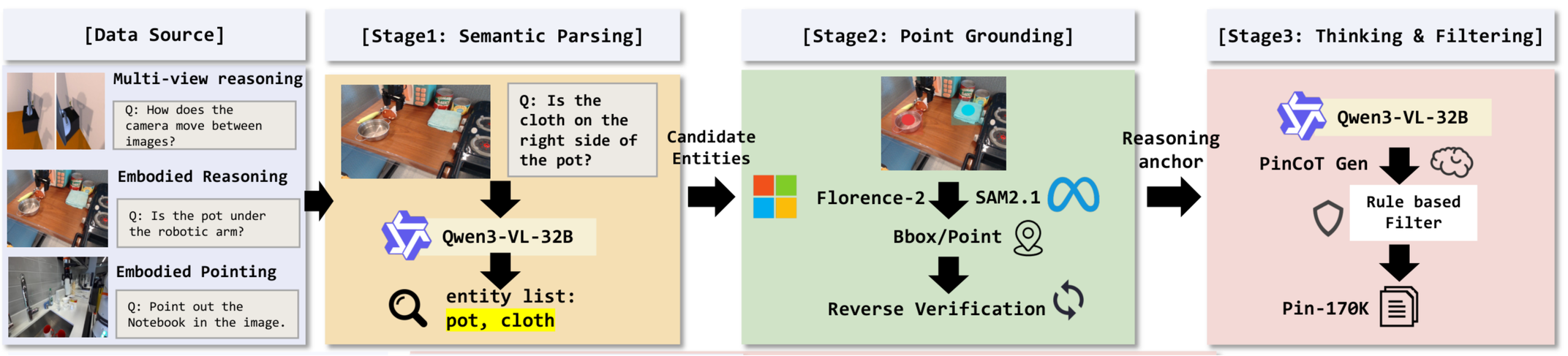
Our insight

Persistent IDs connect visual evidence, intermediate steps and the final answer.

QR PLACEHOLDER

PAPER / CODE

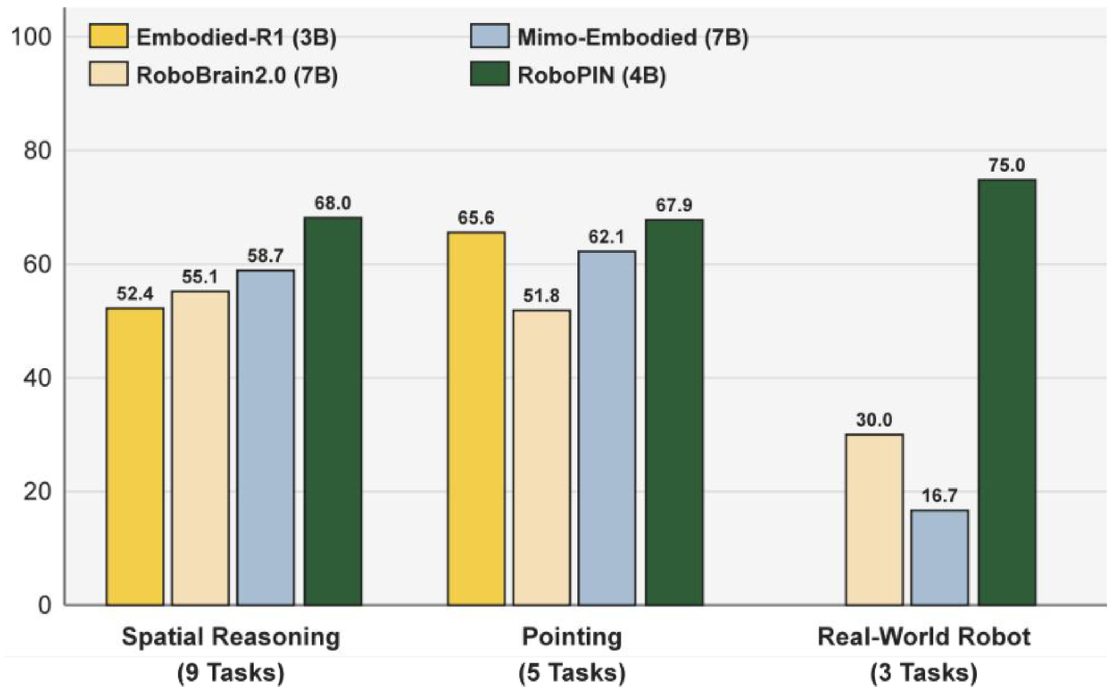
Our Method: RoboPIN / PinCoT



Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

Experimental Results



Ablation studies

Reward design

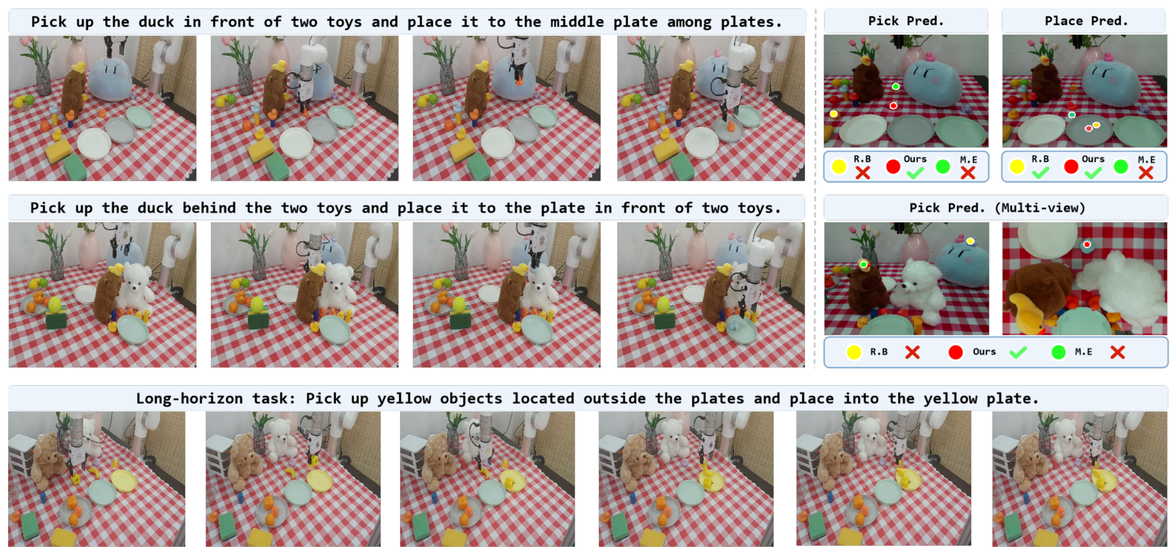
Setting	EmbSpatial
Full reward	**84.7**
w/o Pin	82.0
w/o Consistency	82.7

Training stages

Stage	Spatial	Point
SFT	57.0	66.0
+ CoT-SFT	62.0	63.6
+ RFT	**67.7**	**67.9**

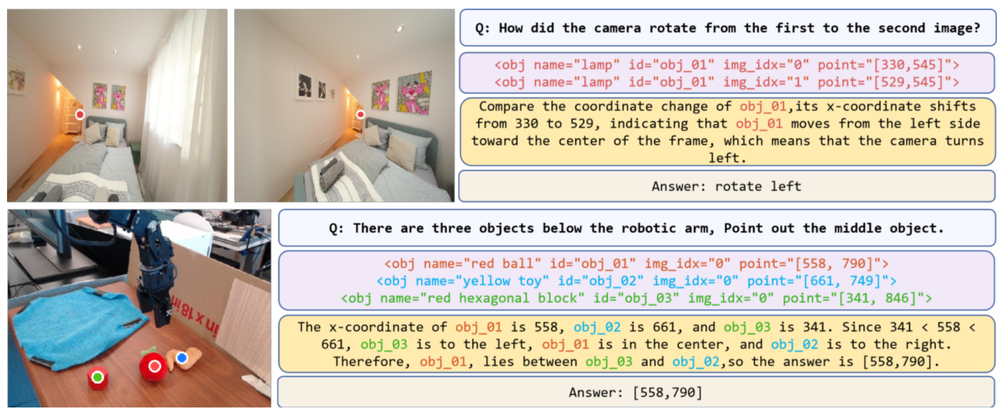
Spatial reasoning: +10.7 points from SFT to RFT.

Qualitative Results



Zero-shot xArm · successes / 20

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon



Persistent visual anchors make embodied reasoning traceable.